FRANMARINE UNDERWATER SERVICES PTY LTD

VESSEL START-UP AND SHUT-DOWN

Procedure

MV INTEGRITY III | UVI: 414562

Marine Order 504 (2024) — Schedule 1 Compliance

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Review Frequency

This document shall be reviewed at least every 12 months, or sooner following a significant incident, change of operations, vessel modification, or change in applicable legislation.

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1. PURPOSE AND SCOPE

This procedure establishes the engine start-up, post-launch verification, and shut-down procedure for MV Integrity III, fitted with twin Suzuki DF175 four-stroke outboard engines running on premium unleaded fuel (no ethanol).

1.1 Regulatory Reference

Marine Order 504 (2024), Schedule 1, Clause 11(5)(a).

1.2 Battery System

MV Integrity III is fitted with two batteries (Battery 1 and Battery 2), each with its own isolator switch. A third battery selector switch provides four positions: OFF, 1, 2, or 1+2 (BOTH). Under normal operations the selector is set to 1+2 so that both batteries share the charging and starting load.

2. POST-LAUNCH CHECKS

Immediately after launching the vessel (or after arriving at the vessel if berthed afloat), the following checks shall be completed before starting engines:

- Confirm the vessel sits well in the water with no unusual list or trim.
- Inspect the vessel for signs of leaks. Check around hull fittings, transom, and bilge area.
- Confirm the bilge system is functioning. MV Integrity III is fitted with 2 x manual bilge pumps (4.0 kL/hr capacity each per NSCV Part G Ch 3 Table 1). Operate the primary pump and verify discharge.

3. ENGINE START-UP PROCEDURE

3.1 Battery Isolation

- Switch both battery isolators to ON.
- Set the battery selector switch to 1+2 (BOTH).
- If one battery is known to be flat, select the good battery (position 1 or 2) for starting, then switch to 1+2 once the engines are running and charging.

3.2 Pre-Start

- Confirm engine kill switch lanyards are attached and functional.
- Confirm outboards are trimmed down to the operating position.
- Confirm gear selector is in NEUTRAL.
- Visually inspect around each outboard for fuel leaks, oil leaks, or debris.

3.3 Starting Engines

- Turn ignition key to the ON position. Allow the fuel injection system to prime (listen for fuel pump).
- Start the port engine. Allow to idle. Immediately check the cooling water tell-tale for steady water flow.
- Start the starboard engine. Allow to idle. Check cooling water tell-tale.
- If either engine does not start on the first attempt, wait 10 seconds before reattempting. If the engine fails to start after three attempts, do not continue — investigate the cause before further attempts.

3.4 Post-Start Verification

With both engines running at idle, verify the following:

- Engine run test: no strange noises, vibrations, or false starts.
- Cooling system is functioning on both engines (steady tell-tale stream from each outboard).
- Oil pressure warning lights are extinguished.
- Battery charging indicators show normal operation on both batteries.
- Steering responds correctly in both directions.
- Throttle and gear selectors operate smoothly through forward, neutral, and reverse.
- Engine gauges (RPM, temperature, voltage) show normal readings.

Allow the engines to warm up at idle for a minimum of 2 minutes before applying load.

4. OPERATIONAL NOTES

- Do not operate engines at sustained high RPM until engine oil temperature has reached operating range.
- Monitor the cooling tell-tale periodically throughout the voyage. Loss of tell-tale flow requires immediate investigation.
- If any engine alarm sounds during operation, reduce throttle to idle, assess the situation, and shut down the affected engine if necessary.
- Operate both engines at matching RPM to maintain balanced thrust and steering.
- Monitor battery voltage on both batteries periodically. If one battery is not charging, investigate the charging circuit before the next voyage.

5. SHUT-DOWN PROCEDURE

5.1 At Berth or Mooring

- Reduce throttle to idle on both engines.
- Place both gear selectors in NEUTRAL.
- Allow engines to idle for 1–2 minutes to allow cooling.
- Turn ignition keys to OFF for both engines.
- Trim outboards up to the stowed position (if the vessel will remain on a mooring or trailer).

5.2 Battery Isolation

If the vessel will be unattended:

- Set the battery selector switch to OFF.
- Switch both battery isolators to OFF.

If the vessel will be attended or on a short stop, the battery selector may remain on 1+2 at the master's discretion.

5.3 Securing the Vessel

- Attach outboard securing straps (if applicable).
- Close all hatches, lockers, and access points.
- Remove or secure all loose equipment.
- Deploy fenders and mooring lines as appropriate (refer FUS-SMS-INT3-05C).
- Check bilge is dry. If water is present, identify the source before leaving the vessel.

6. RECORDS

Engine start-up and shut-down checks are captured as part of the QR code pre-start form and the physical vessel logbook. Any defects or anomalies identified during start-up or shut-down shall be recorded and reported to the East Coast Operations Superintendent.

7. REFERENCES

- FUS-SMS-INT3-05A Pre-Departure Checks
- FUS-SMS-INT3-07 Maintenance
- Suzuki DF175 Owner’s Manual
- NSCV Part G Ch 3 Table 1 (Bilge Pumps)
- Marine Order 504 (2024), Schedule 1, Clause 11(5)(a)